

10-606 Recitation

Justin

September 2026

Warmup

Problem 1

Consider the ceiling function $f(x) = \lceil x \rceil$. For any real number, $f(x)$ rounds x up to the nearest whole number. Write a mathematical description of the type of function f . Then write python code that implements ceiling as a lambda function.

Solution: $\mathbb{R} \rightarrow \mathbb{Z}$

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f = lambda x: x if int(x) == x else int(x) + 1
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Problem 2

Write a set-builder notation that finds elements of $\mathbb{N}$ between 1 and 80 that are not divisible by 2, 3, or 7.

Solution: $\{x \in \mathbb{N} \mid 1 \leq x \leq 80 \mid 2 \nmid x, 3 \nmid x, 7 \nmid x\}$

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{x for x in range(1, 81) if x % 2 != 0 and x % 3 != 0 and x % 7 != 0}
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Matrix Applications and Properties

Problem 1: System of Equations in Matrix Form (3x3)

Consider the following system of equations:

$$2x + y - z = 8$$

$$3x - 2y + 4z = -2$$

$$x + y + z = 4$$

Write this system in matrix form $A\mathbf{x} = \mathbf{b}$ and discuss how the solution can be found using the inverse of A .

Solution: We can write the system in matrix form as:

$$\begin{bmatrix} 2 & 1 & -1 \\ 3 & -2 & 4 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 4 \end{bmatrix}$$

This system can be solved by computing the inverse of A , i.e.,

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

Problem 2: Matrix Multiplication (3x3)

Let

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}.$$

Compute AB and BA . What do you observe?

Solution: Notice that

$$AB \neq BA,$$

demonstrating that matrix multiplication is not commutative.

Problem 3: Associative Property of Matrix Multiplication (3x3)

Verify the associative property of matrix multiplication by calculating $(AB)C$ and $A(BC)$, where

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{bmatrix}.$$

Solution: First, compute AB :

$$AB = \begin{bmatrix} 30 & 24 & 18 \\ 84 & 69 & 54 \\ 138 & 114 & 90 \end{bmatrix}.$$

Now compute $(AB)C$:

$$(AB)C = \begin{bmatrix} 372 & 300 & 228 \\ 1020 & 822 & 624 \\ 1668 & 1344 & 1020 \end{bmatrix}.$$

Now compute BC :

$$BC = \begin{bmatrix} 150 & 120 & 90 \\ 96 & 78 & 60 \\ 42 & 36 & 30 \end{bmatrix}.$$

Then:

$$A(BC) = \begin{bmatrix} 372 & 300 & 228 \\ 1020 & 822 & 624 \\ 1668 & 1344 & 1020 \end{bmatrix}.$$

Thus,

$$(AB)C = A(BC),$$

demonstrating the associative property.

Problem 4: Distributive Property of Matrix Multiplication (3x3)

Let

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Verify the distributive property by calculating $A(B + C)$ and $AB + AC$.

Solution: First, compute $B + C$:

$$B + C = \begin{bmatrix} 4 & 2 & 2 \\ 6 & 6 & 4 \\ 10 & 8 & 8 \end{bmatrix}.$$

Now compute $A(B + C)$:

$$A(B + C) = \begin{bmatrix} 98 & 82 & 74 \\ 118 & 98 & 88 \\ 138 & 114 & 102 \end{bmatrix}.$$

Now compute AB :

$$AB = \begin{bmatrix} 90 & 78 & 66 \\ 108 & 93 & 78 \\ 126 & 108 & 90 \end{bmatrix}.$$

Compute AC :

$$AC = \begin{bmatrix} 8 & 4 & 8 \\ 10 & 5 & 10 \\ 12 & 6 & 12 \end{bmatrix}.$$

Therefore,

$$AB + AC = \begin{bmatrix} 98 & 82 & 74 \\ 118 & 98 & 88 \\ 138 & 114 & 102 \end{bmatrix}.$$

Hence,

$$A(B + C) = AB + AC.$$

Norms and Their Applications

Problem 1: Manhattan Norm ($\|\mathbf{x}\|_1$)

Consider the vector

$$\mathbf{x} = \begin{bmatrix} 3 \\ -4 \\ 5 \end{bmatrix}.$$

Calculate the $\|\mathbf{x}\|_1$ (Manhattan norm) of the vector.

Solution: The Manhattan norm (or ℓ_1 -norm) is defined as the sum of the absolute values of the vector components:

$$\|\mathbf{x}\|_1 = |3| + |-4| + |5| = 3 + 4 + 5 = 12.$$

Problem 2: Euclidean Norm ($\|\mathbf{x}\|_2$)

Given the vector

$$\mathbf{y} = \begin{bmatrix} 6 \\ 8 \\ -2 \end{bmatrix},$$

calculate the Euclidean norm $\|\mathbf{y}\|_2$.

Solution: The Euclidean norm (or ℓ_2 -norm) is the standard length of the vector and is computed as:

$$\|\mathbf{y}\|_2 = \sqrt{6^2 + 8^2 + (-2)^2} = \sqrt{36 + 64 + 4} = \sqrt{104} \approx 10.2.$$

Problem 3: Infinity Norm ($\|\mathbf{x}\|_\infty$)

For the vector

$$\mathbf{z} = \begin{bmatrix} -10 \\ 5 \\ 7 \end{bmatrix},$$

calculate the infinity norm $\|\mathbf{z}\|_\infty$.

Solution: The infinity norm (or ℓ_∞ -norm) is defined as the maximum absolute value of the vector components:

$$\|\mathbf{z}\|_\infty = \max(|-10|, |5|, |7|) = 10.$$

Problem 4: General p -Norm

Given the vector

$$\mathbf{w} = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}$$

and $p = 3$, calculate the $\|\mathbf{w}\|_3$.

Solution: The p -norm is defined as:

$$\|\mathbf{w}\|_p = \left(\sum_{i=1}^n |w_i|^p \right)^{1/p}.$$

For $p = 3$, we have:

$$\begin{aligned}\|\mathbf{w}\|_3 &= (|1|^3 + |-3|^3 + |4|^3)^{1/3} \\ &= (1 + 27 + 64)^{1/3} = 92^{1/3} \approx 4.51.\end{aligned}$$

Problem 5: Application in Machine Learning - Frobenius Norm

Consider the matrix

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & -9 \end{bmatrix}.$$

Compute the Frobenius norm $\|A\|_F$.

Solution: The Frobenius norm is a matrix norm and is defined as the square root of the sum of the squares of all matrix elements:

$$\begin{aligned}\|A\|_F &= \sqrt{2^2 + (-1)^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + (-9)^2} \\ &= \sqrt{4 + 1 + 9 + 16 + 25 + 36 + 49 + 64 + 81} = \sqrt{285} \approx 16.88.\end{aligned}$$

Vector Space Basics

Problem 1: What is / isn't a vector space

1. True or false: the set of all 2×2 invertible matrices with real entries is a vector space under usual addition and scalar multiplication.

Solution: False. It is not closed under addition (and the zero matrix is not invertible).

2. Is the set of polynomials of degree exactly 3 (not at most 3) a vector space? Why or why not?

Solution: No. It is not closed under addition. For example,

$$x^3 + (-x^3) = 0,$$

and the zero polynomial is not a polynomial of degree 3.

3. Is

$$\{x \in \mathbb{R} \mid x^2 \leq 9\}$$

a vector space under pointwise addition and scalar multiplication?

Solution: No. The set can be simplified to

$$\{x \in \mathbb{R} \mid -3 \leq x \leq 3\}$$

But, if we take a constant such as 4 , $4 * 3$ is outside the set.

4. Let $A \in \mathbb{R}^{n \times m}$ be a real-valued matrix. Is the set of all vectors $x \in \mathbb{R}^m$ such that $Ax = 0$ a vector space (under usual addition and scalar multiplication)?

Solution: Yes. The zero vector is in the set, and the set is closed under scalar multiplication and addition.

Problem 2: Linear Dependence & Span

1. Are the vectors

$$(1, 1, 0), \quad (0, 1, 1), \quad (1, 2, 1)$$

linearly dependent or independent?

Solution: They are linearly dependent because

$$(1, 2, 1) = (1, 1, 0) + (0, 1, 1).$$

2. Does

$$\{(1, 2, 3), (2, 4, 6), (0, 1, 0)\}$$

span $\mathbb{R}^3$?

Solution: No. The first two vectors are multiples of each other:

$$(2, 4, 6) = 2(1, 2, 3).$$

Therefore, together with $(0, 1, 0)$, they span at most a two-dimensional subspace.